

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Assignment # 14

Date Given: November 26, 2025

Date Due: December 17, 2025
Total Points: 10

1. Suppose $a_n \rightarrow \lambda > 0$ as $n \rightarrow \infty$, then show that

$$\left(1 + \frac{a_n}{n}\right)^n \rightarrow e^\lambda.$$

Hence, or otherwise show that for any $t \in \mathbb{R}$, *bold*,

$$\left(1 - \frac{t^2/2}{n} + o\left(\frac{1}{n}\right)\right)^n \rightarrow e^{-t^2/2}.$$

2. Find the *characteristic function* of the χ_n^2 -distribution.
3. Show that if X is symmetric around then its characteristic function is real valued.
4. Show that by using a DCT argument that characteristic function is *continuous*.